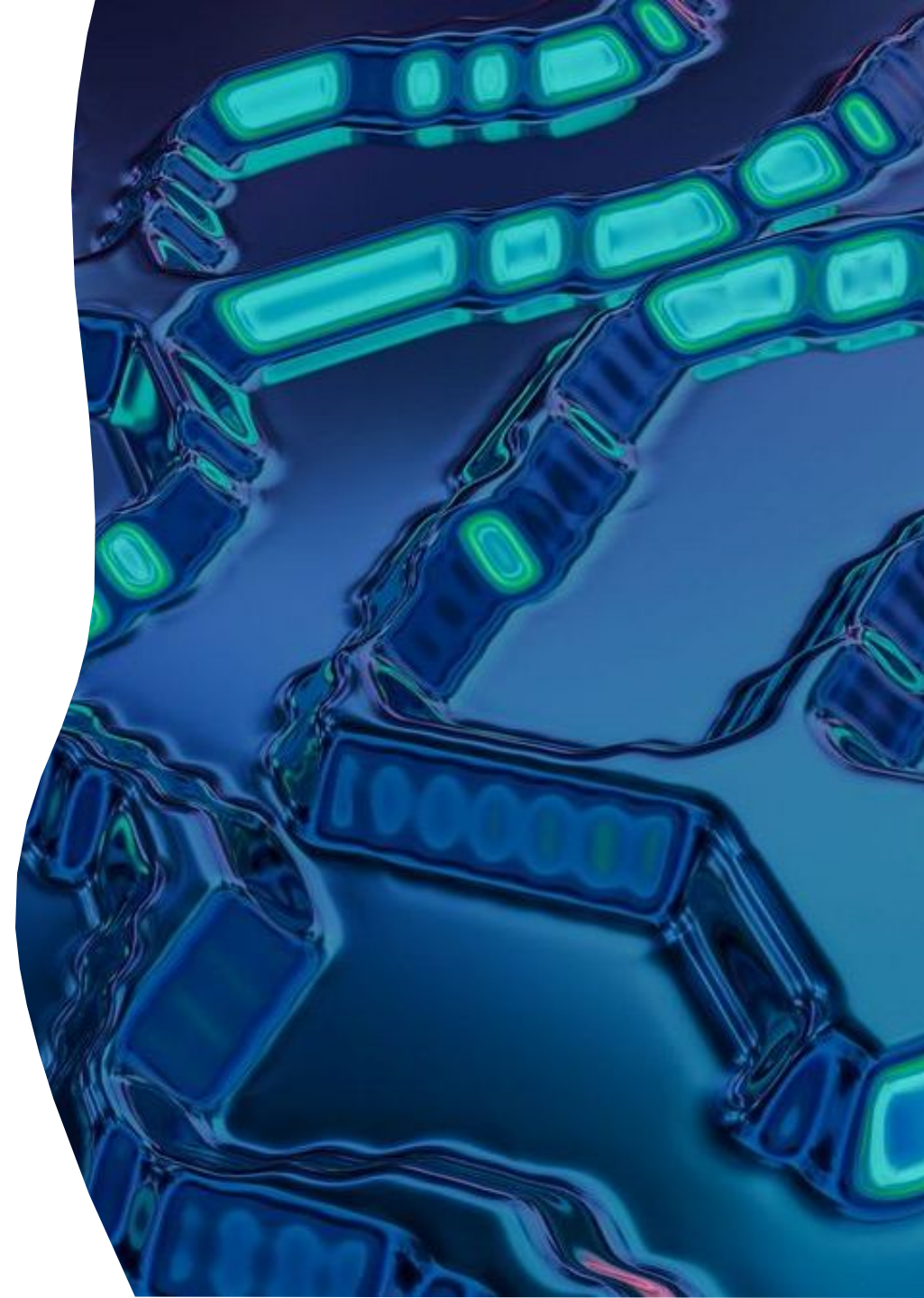


Data Analytics, Data Mining and Machine Learning

Lecture 2

Narges Chinichian · Semester 1, 2026



Overview of this course

Week	Date	Topic
1	07.09.2026	Introduction
2	14.09.2026	Data, Research Questions & Ethics
3	21.09.2026	Data Visualisation & Exploratory Data Analysis
4	28.09.2026	Preprocessing & Feature Engineering
5	05.10.2026	Machine Learning Fundamentals & Regression
6	12.10.2026	Project Presentations I
7	19.10.2026	Classification: Logistic Regression, KNN & Decision Trees
8	26.10.2026	Bank Holiday — No Class
9	02.11.2026	Gradient Descent, Regularisation & Model Complexity
10	09.11.2026	Model Evaluation & Ensemble Methods
11	16.11.2026	Neural Networks, Unsupervised Learning & Special Topics
12	23.11.2026	Project Presentations II

Quick Quiz

Check Loop

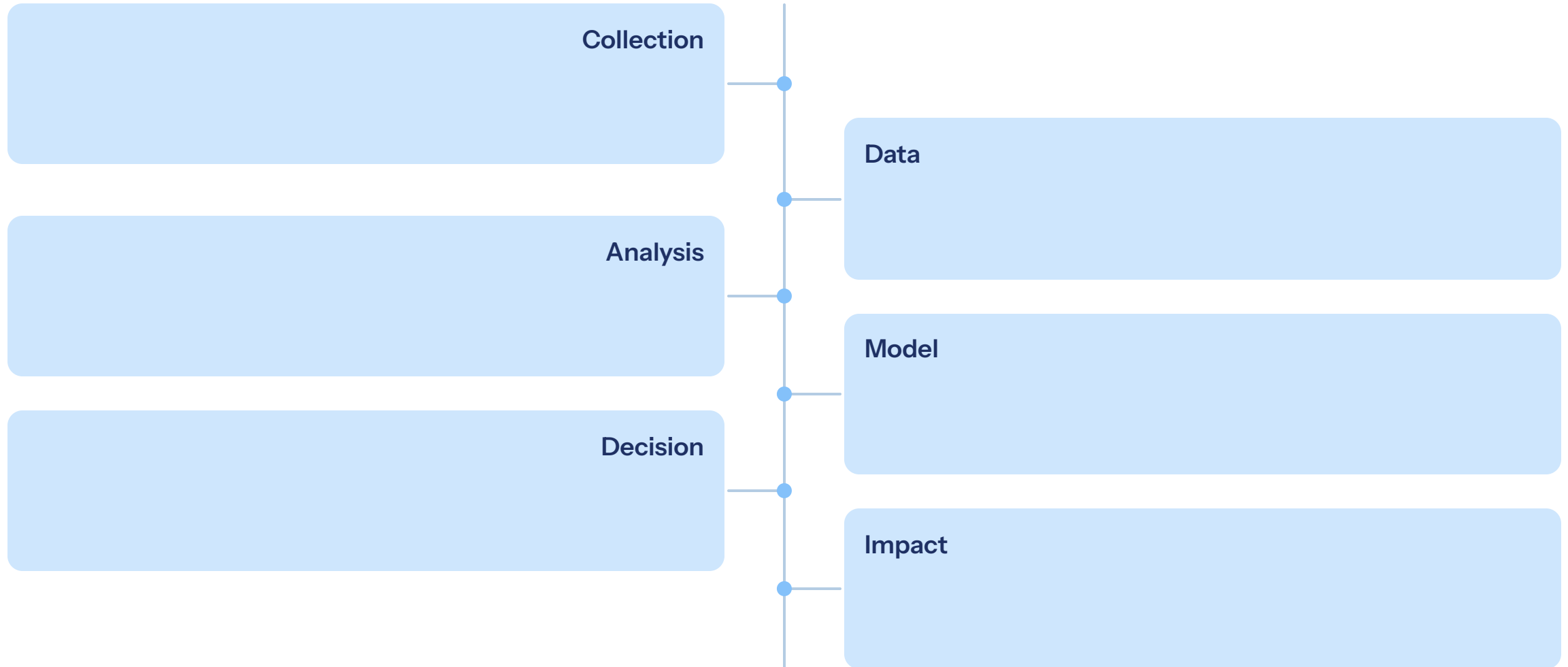




Today's Agenda

- Group and research questions being discussed
- Data Ethics

Ethical problems can enter at every stage of a project



We will walk this pipeline one stage at a time. Every case today is pinned to the stage where the problem first entered.

At each stage, ask:

What could go wrong?

Let's have a look at a few examples from David Dao et al., "Awful AI": curated collection of problematic AI uses. DOI [10.5281/zenodo.5855972](https://doi.org/10.5281/zenodo.5855972)

Legal $\neq$ ethical

Problematic cases can be:

Legal but ethically questionable

Ethically desirable but legally restricted

Clearly illegal and unethical

Do not treat **"It's legal."** as the end of an ethical discussion.

Clearview AI

Clearview AI built a facial-recognition database by collecting photographs from the web and social-media profiles.

The technical question is:

Can we collect these images?

But the ethical question is:

Should we?

And:

Would the people in the images reasonably expect their photographs to be used this way?

⚠ Publicly accessible ≠ ethically unrestricted

Source: David Dao et al., "Awful AI", "Clearview.ai". Read more: [The New York Times](#)



Cambridge Analytica

Data collected through Facebook was used in ways that raised major questions about political profiling and behavioural influence.

The important question is not simply:

"Did someone provide the data?"

It is:

"What did they provide it for?"

📄 Purpose matters.

Data collected for one purpose does not automatically become ethically appropriate for every other purpose.

Source: David Dao et al., "Awful AI", "Cambridge Analytica". Original sources cited by Awful AI: [The Guardian](#).



Consent is not a blank cheque

Imagine:

"I agree to share my data to use this fitness app."

Does that automatically mean:

- *"You can sell my data to an insurer."*
- *"You can use it to predict my health risks."*
- *"You can use it to train a completely unrelated AI system."*



Ask: what exactly did the person consent to?

General ethical framework; purpose and consent issues are illustrated by cases in David Dao et al., "Awful AI".

Who is represented in the data?

A dermatology AI application was reported to have been trained using a dataset in which only **3.5% of images came from people with darker skin**.

What could happen?

- The system may perform differently across populations.
- A high overall accuracy could hide poor performance for some groups.
- The people most affected may be those least represented in the training data.

Ask yourself

Who is missing from the dataset?

Source: David Dao et al., "Awful AI", "Dermatology App". Original source cited by Awful AI: Read on [VICE](#).

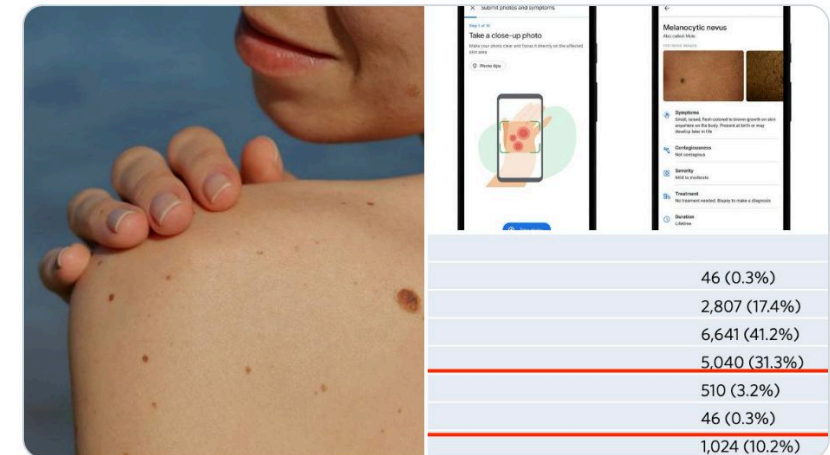


Ade Adamson, MD MPP

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Google launches AI health tool for skin conditions in Europe on.ft.com/3tWNQtS The algorithm was developed based on training data with less than 4% dark skin types. It should come with a warning BEWARE OF RESULTS IF BLACK!!!



10:07 PM · May 18, 2021



189



Reply



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STAGE 2 — DATA

Amazon's recruitment algorithm

Amazon developed an AI system to help evaluate job applicants.

The system was trained using historical hiring data.

The historical data reflected a workforce in which men were overrepresented.

The system learned patterns associated with previous hiring decisions and reportedly penalised CVs containing terms such as "**women's chess club captain.**"

Ask yourself

If historical decisions contain bias, what happens when we train a model to reproduce them?

Source: David Dao et al., "Awful AI", "Sexist Recruiting". Read more: [Reuters](#)



Bias enters before the model

A common assumption: *biased data* → *biased model*. But the data is already the output of at least three decisions.

Who is included?

Who dropped out, never enrolled, could not access the survey, did not respond, or was never in the sampling frame?
The people missing from a dataset can matter as much as the people inside it.

How was it labelled?

Who decided what each label means, and who applied it?

What was measured?

A variable is not a fact about the world — it is the result of a measurement process. What exactly was measured, how, by whom, and could that process differ between groups?

The model may simply learn what the dataset contains.

What can we infer about someone?

A dataset does not need to contain **sexual orientation, health condition, political beliefs** or anything else sensitive for an analysis to reveal information about these things.

The *Awful AI* collection includes examples of systems claiming to infer sensitive characteristics from facial images, including sexual orientation and genetic disease.

This creates a different ethical question:

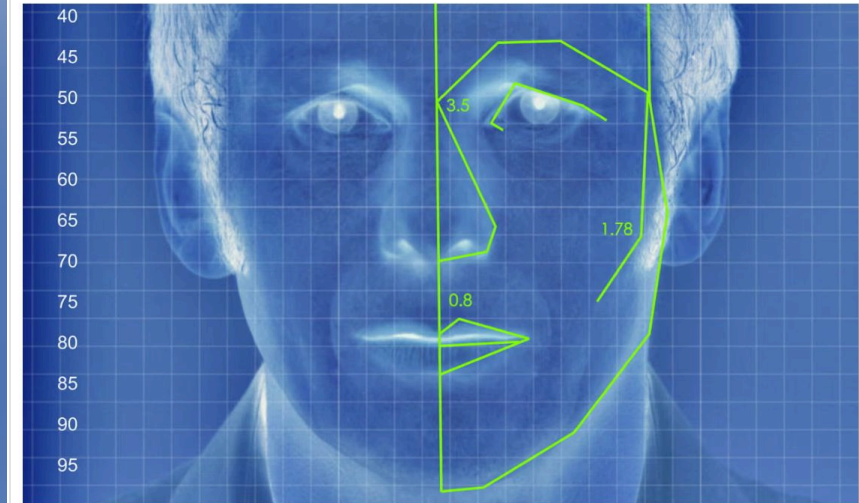
Is it acceptable to predict something about someone that they never chose to disclose?

Source: David Dao et al., "Awful AI", "[AI-based Gaydar](#)" and "Infer Genetic Disease From Face".

Source image: [The Guardian](#), "[New AI can guess whether you're gay or straight from a photograph](#)" (2017).

New AI can guess whether you're gay or straight from a photograph

An algorithm deduced the sexuality of people on a dating site with up to 91% accuracy, raising tricky ethical questions



📷 An illustrated depiction of facial analysis technology similar to that used in the experiment.
Illustration: Alamy

Let's watch this together



TED TED

Leo Anthony Celi: Data Bias is the Waterloo of Health AI



Data can reveal more than it contains.

Imagine a dataset containing age, location, browsing behavior, purchases and social-media activity

None of the columns says "**mental health**".

But an algorithm may still find patterns associated with mental health.



Data ethics therefore concerns **collection + inference** — not just what is explicitly stored.

Accuracy is not the whole story

Model A — 95% accuracy

99% for Group A · 70% for Group B

Model B — 93% accuracy

94% for Group A · 92% for Group B

Which is better? Which would you choose?

This is the dermatology case in numbers: **Model A is the system whose high overall accuracy hides poor performance for the least-represented group.**

There may not be one universally correct answer. But you need to be able to justify the choice.

When analysis changes behaviour

Ask:

Who benefits from the optimisation?

When does personalisation become manipulation?



TED TED



Tristan Harris: How a handful of tec...

A handful of people working at a handful of tech companies steer the thoughts of...

Tristan Harris: *How a handful of tech companies control billions of minds every day*

Source: David Dao et al. and "[Attention Engineering](#)"

Who benefits? Who bears the risk?

Suppose an algorithm predicts whether someone is likely to: receive a loan, get hired, drop out of university, commit fraud or need healthcare

Ask two questions:

Who benefits if the prediction works?

Who bears the cost if it is wrong?

The person building the model may gain **+1% accuracy**, while the person being classified may experience **"I didn't get the loan."**

Ownership & credit

In September 2026, OpenAI claimed that an internal AI system had solved the Navier–Stokes Millennium Prize problem. Mathematician Tristan Buckmaster and Anthropic researcher Levent Alpöge raised concerns after working on closely related research using OpenAI's tools.

- Their research was unpublished
- Their work was stored in OpenAI's Codex
- OpenAI says it did not access specific user data
- OpenAI also said it could not rule out that de-identified user data had helped improve its models
- The researchers disputed how credit and authorship were handled

Technical question:

Can an AI system access or learn from data users provide to it?

Ethical question:

Should researchers' unpublished work be used to advance someone else's research?

Legal question:

What rights do researchers have over work submitted to AI tools?

📌 Access $\neq$ permission $\neq$ credit

Sources: Financial Times, "OpenAI faces competing claims around maths breakthrough" (2026); ABC News, "Controversy erupts as OpenAI claims solution to Navier Stokes maths problem" (2026).

A data-ethics checklist

One question per stage. Same pipeline, now as a working tool.

01

Collection

Why was this data collected, and would the people in it expect this use?

03

Analysis

Could sensitive information be inferred, even if it was never stored?

05

Decision

What decisions could be made from this analysis?

02

Data

Who is represented, who is missing, and what does each important variable actually measure?

04

Model

Does overall performance hide failure for any group?

06

Impact

Who benefits, and who bears the cost if we are wrong?

Your project: the ethics check

- 1 Where did the data come from?
- 2 Why was it originally collected?
- 3 Who is represented?
- 4 Who might be missing?
- 5 What personal or sensitive information does it contain? Explicitly or implicitly?
- 6 What could your analysis reveal that was not obvious in the original data?
- 7 Who could benefit from your findings?
- 8 Who could be harmed or misrepresented?

☐ Data ethics is not a final checkbox at the end of a project. Ethical questions exist at every step of the pipeline.

The question to remember

Don't only ask:

"Can I analyse this data?"

Also ask:

"What happens if I do?"

And:

"Who bears the consequences?"